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How does a poor labour market affect inequalities in access to postsecondary education? Empirical evidence from 31 affluent countries

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Abstract

This study examines how macroeconomic conditions affect inequality of educational opportunity. Using longitudinal data from 30 European countries and the USA, we study the effect of changing unemployment rates on the extent of social inequality in postsecondary transitions in the period (2004–2016), which includes the Great Recession. The findings from multilevel models with country fixed effects show that rising unemployment rates tend to increase social inequalities in postsecondary transitions, as youth from low-educated families have increased risk for exclusion from both employment and education, while the likelihood of postsecondary enrolment slightly rises for youth from higher-educated families. Yet, economic downturns also incentivize youth whose parents have middle level of education to opt for postsecondary education, especially in contexts where it is more affordable. Our findings suggest that changing opportunity costs and economic insecurity are the relevant factors to explain changing inequalities in access to postsecondary education during a recession.

Key words: education; inequality; public policy; social mobility; unemployment

JEL classification: I240 Education and Inequality, J640 Unemployment: Models, Duration, Incidence, and Job Search, I220 Educational Finance, Financial Aid

1. Introduction

A significant body of research has shown that inequalities in tertiary education attainment vary across countries and change over time depending on institutional arrangements. The main focus of this research has been on the role of education systems, educational reforms,

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and expansion in affecting inequality of educational opportunity (e.g. Raftery and Hout, 1993; Breen and Jonsson, 2005; Arum *et al.*, 2007; Triventi, 2013; Bernardi and Ballarino, 2014; Pöyliö *et al.*, 2018). However, the Great Recession and its consequences on families with children (e.g. Cantillon *et al.*, 2017; Garfinkel *et al.*, 2016) have raised interest in the effects of economic downturns on the inequality in educational attainment.

Overall, we know relatively little about how macroeconomic conditions affect inequality of educational opportunity and how they drive the intergenerational effects of social background (Torche, 2010; Macmillan, 2014). On the one hand, we can expect social inequality in educational transitions to increase during a recession. Economic downturns disproportionally affect the unemployment risks and wages of less advantaged families, thus resulting in a greater relative decline in their resources and economic security (Dellas and Sakellaris, 2003). Hence, it might be more difficult for these families to invest in their children's education, while privileged parents can increase their investments when faced with an uncertain economic context (Mayer, 2001; Lunn and Kornrich, 2018). On the other hand, it would also be possible to think that social inequality in postsecondary enrolment is declining during a recession. As increasing unemployment rates and declining opportunities in the labour market will imply lower opportunity costs of pursuing postsecondary education during a recession, an increasing number of young people from less advantaged families might enrol, while aspiration for status maintenance might motivate highly educated families to invest in postsecondary education regardless of the macroeconomic context (Bozick, 2009).

Previous research on the relationship between economic downturns and inequality in postsecondary enrolment is rare, has focused on single countries and has provided mixed evidence: some single-country studies suggest that social inequalities in enrolment increase during economic downturns (Christian, 2006), whereas others find the opposite (Sievertsen, 2016). A notable exception is a comparative study of 24 countries by Salazar *et al.* (2020) that used the Trends In International Mathematics And Science Study (TIMSS) data for the years 2003, 2007 and 2011 and showed that the Great Recession increased inequality in educational expectations between eighth graders from different social backgrounds. However, the connection between educational expectations and attainment is complex. Many students change their expectations throughout their school years and youth with volatile expectations are less likely to enrol in postsecondary education compared to those with more stable expectations (Bozick *et al.*, 2010). Therefore, further comparative research is essential to explore whether recessions may systematically increase inequalities in actual enrolment behaviour.

We can expect economic recessions to affect postsecondary enrolment due to individual-level changes in a family's economic resources or perceived financial security as well as due to macro-level changes such as reduced opportunities in the labour market for young people. Importantly, institutional arrangements might moderate the negative effects of economic downturns on inequalities in educational transitions. In particular, when we consider educational decision-making as a process based on perceived costs and benefits (Breen and Goldthorpe, 1997), the affordability of tertiary education is likely to affect the range of available transition alternatives. Thus, it may help prevent disadvantageous school-to-work transitions during an economic recession, even or maybe especially among children from less privileged social backgrounds.

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In this study, we therefore examine how changing unemployment rates affect the extent of social inequality in postsecondary educational transitions using longitudinal data from 30 European countries and the USA. We focus on transitions that young people make after completing upper secondary school, such as enrolment in postsecondary education, entry to employment or not in education, employment or training (NEET). Our analysis covers transitions in the years 2004–2016, a period that includes the Great Recession when unemployment rates rose and wages declined in many Western countries. This resulted in increasing child poverty and greater difficulties in school-to-work transitions for young people (Cantillon *et al.*, 2017; Schoon and Bynner, 2019), but also in increased average enrolment rates in postsecondary education (e.g. Barr and Turner, 2013). However, there is significant variance across countries and over time in unemployment rates. We consider this variance to explore how macroeconomic conditions affect the extent of social inequality in youth transitions.

We focus on the unemployment rate as a key indicator of macroeconomic troubles and limited economic prospects for young people and their families. In contrast to using measures for gross domestic product (GDP) and its growth (e.g. Salazar *et al.*, 2020) or another indicator that is more closely related to a technical macroeconomic definition of a recession, our focus is grounded in the assumption that labour markets are the main source of economic livelihood for the very broad majority of families and directly influence opportunities to enter the job market after completing secondary school. We further believe that adopting this focus is especially relevant in international comparisons, as some countries are more successful in decoupling GDP declines and actual difficulties in the labour market, while others have little institutional protection available to shield jobs during economic downturns (see also Redbird and Grusky, 2016). Moreover, instead of defining a recession as a dummy variable, we chose to use a quantitative measure (unemployment rate) to capture more fine-grained differences across countries in the severity and timing of poor labour market conditions and in the gradual recovery.

In our empirical analysis, we also assess how much the changing economic resources of families explain why recessions amplify or reduce inequalities in postsecondary transitions. In addition, we explore the moderating effect of the affordability of tertiary education. Our analysis is based on data from five longitudinal studies: the European Union Statistics on Income and Living Conditions (EU-SILC), the US Survey of Income and Program Participation (SIPP), the German Socio-Economic Panel (SOEP), the British Household Panel Survey (BHPS) and its successor Understanding Society (UKHLS). We use multilevel regression models with country fixed effects to first identify the effect of changing economic conditions and then estimate its relevance for youth with different social backgrounds.

Empirically, our study provides new insights from a comparative perspective, revealing that a rising level of unemployment somewhat increases social inequalities in enrolment to postsecondary education. We observe a slight rise in enrolment rates for youth from advantaged families with increasing unemployment, but there is almost no change for their peers from low-educated families, who face an elevated risk of staying out of employment and education. Our research also contributes novel empirical evidence on the role of family economic resources and the affordability of tertiary education in shaping inequalities during economic downturns. We find that youth whose parents have a middle level of education (upper secondary) increasingly participate in postsecondary education in response to rising unemployment rates, especially in contexts where tertiary education is more affordable.

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2. Theoretical background

2.1 The effects of recessions

Economic recessions can have decisive consequences on the life courses of young people by affecting their participation in education and the labour market. Recessions are likely to put psychological and financial burdens on families. In particular, economic adversity could affect the educational attainment through lower investments in children, less sensitive parenting practices and poorer health (e.g. the review in Duncan *et al.*, 2018). For instance, the family stress model (Conger *et al.*, 1992) expects economic hardship during recessions to harm the quality of family relationships, parenting and child's development (cf Garfinkel *et al.*, 2016).

Yet extensive prior research, mostly done in economics, has suggested that investing in education is countercyclical to the business cycle (e.g. Clark, 2011; Barr and Turner, 2013; Sievertsen, 2016). Potential reasons relate to lower opportunity costs of education in terms of wage offers during economic recessions (Dellas and Sakellaris, 2003) and students' preferences to wait for the improvement of the economy before starting a job search or their higher incentives to secure a better starting position in the labour market through a higher level of education (Gambetta, 1987; Bozick, 2009).

However, the effect of macroeconomic conditions on youth postsecondary transitions might vary by students' social backgrounds. In the following, we introduce a respective theoretical framework for this study.

2.2 Economic resources and uncertainty

Limited financial resources might make tertiary education entry during an economic recession more difficult for young people from lower socioeconomic backgrounds. Their families typically suffer more in recessions as unemployment rates and wages of low-skilled workers tend to be more sensitive to the business cycle (Dellas and Sakellaris, 2003). Hence, also the losses of wealth during recessions are disproportionally concentrated among less well-off households (Pfeffer *et al.*, 2013). In addition, students from low-income families are more likely to have to work to finance their studies (Bozick, 2007) but finding part-time employment is more difficult in periods of high unemployment.

Beyond the direct impact on financial resources, an economic recession might raise perceptions of uncertainty about families' capability to invest in education as well as lower anticipated returns to education. Anticipated returns likely depend on the business cycle (Dellas and Koubi, 2003), as graduating during a recession has long-term negative effects on earnings and professional employment opportunities (Kahn, 2010). The lower returns to tertiary education might trigger more risk-averse transition behaviour, particularly among students from less advantaged families whose educational decisions tend to be more affected by risk aversion (Erikson and Jonsson, 1996; Breen *et al.*, 2014). In other words, their decisions might be more affected by the perception of tertiary education as a long-term commitment with uncertain outcomes and this could be aggravated by a recession.

Some previous studies support these considerations. More specifically, two studies using data from the USA found that enrolment rates of adolescents from low-income families are adversely affected by increases in national unemployment rates (Christian, 2006) and in state-level job losses (Ananat *et al.*, 2017). Moreover, Salazar *et al.* (2020) found in a comparative study of 24 countries that the Great Recession lowered educational expectations

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more among youth from low-educated families compared to those from high-educated families.

Besides affecting the choice between postsecondary enrolment and labour market entry, severe difficulties at making the transition might lead to exclusion from both employment and education. Above all, adolescents from lower socioeconomic backgrounds might fail to find work and be unable to afford postsecondary education when facing the economic recession (Kalil, 2013). Previous studies in the UK and Germany have shown that when local unemployment rates increase, then young people from lower socioeconomic backgrounds or from unemployed households struggle more to find a job compared to their similarly qualified peers (Macmillan, 2014; Zwysen, 2016), possibly due to limited beneficial social networks and other family resources. Hence, when the unemployment rate increases, young people from disadvantaged backgrounds might struggle to enter postsecondary education or the labour market, and may instead end up in inactivity.

Adverse macroeconomic conditions might also affect the transition behaviour of advantaged families. Although not much is known about the effects of recessions, research focusing on rising income inequality as an adverse macroeconomic condition has proposed that rising inequalities could motivate parents to provide more extensive safety nets to their children by increasing investments in education (Mayer, 2001). More specifically, high-income families might feel more pressure to invest in their children in circumstances where staying behind is risky in terms of future financial prospects (Kornrich and Furstenberg, 2013). Hence, it is possible that recessions motivate especially better off parents to provide an extensive safety net for their children by increasing investments in education, while less advantaged parents struggle to keep up with them. For instance, Lunn and Kornrich (2018) found that in the USA, high-income families increased their spending on (precollege) education during the Great Recession, which resulted in the growing spending difference between high-income and low-income families.

Based on uncertainties and a resource perspective, we propose an *economic resources hypothesis (H1)* expecting that rising unemployment rates lead to the greatest increase in postsecondary enrolment for youth from tertiary-educated families, followed by those whose parents have upper secondary education, while enrolment likely decreases for youth whose parents have low levels of education.

2.3 Opportunity costs and status maintenance

Sociological approaches emphasize that the socioeconomic status of parents reflects not only their socioeconomic resources, but also various other aspects including educational expectations and status aspirations (Haller and Portes, 1973; Andrew and Hauser, 2011). Some previous empirical research has suggested that children of highly educated parents have stable expectations for higher education attendance while their peers from lower socioeconomic backgrounds might be less likely to hold on to their high expectations (Alexander *et al.*, 2008; Bozick *et al.*, 2010; Johnson and Reynolds, 2013). Moreover, the relative risk aversion principle in educational choice models assumes that families seek foremost to avoid downward mobility and thus, aim to make educational decisions that allow children to achieve a class position at least as good as that of their parents (Breen and Goldthorpe, 1997). Hence, adolescents from advantaged families might opt for tertiary education regardless of the macroeconomic conditions as advantaged families are likely to have

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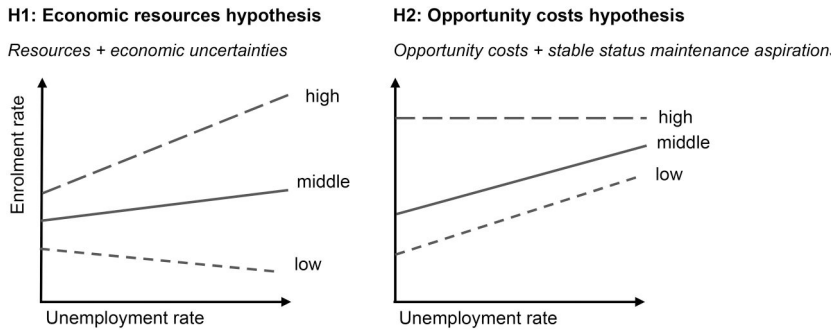

Figure 1. Hypotheses: how the effect of unemployment rate on enrolment depends on education of parents.

Notes: High—parent has tertiary education; middle—upper secondary education (high school); low—lower secondary or less.

a strong aspiration for their children to attain tertiary education in order to maintain social status.

However, young people from lower-educated families could achieve social status comparable to their parents also without postsecondary education and thus their choice might be more affected by opportunity costs. For instance, using Italian data, Barone *et al.* (2018) showed that perceived opportunity costs associated with attending university are relevant in explaining social inequalities in enrolment. Importantly, opportunity costs depend on the macroeconomic context. When unemployment rates are low and more jobs are available, adolescents from less advantaged families might be more likely to prefer immediate income from work over a long-term investment in education compared to their peers from privileged backgrounds (Bozick, 2009). Moreover, high unemployment rates are likely to disproportionally affect the availability of less skilled jobs, potentially encouraging students to stay in postsecondary education when opportunities in the labour market are limited (e.g. see the discussion of the parking theory in Gambetta, 1987). For instance, some previous research has found that enrolment decisions of young people from disadvantaged backgrounds are more responsive to business cycles and local labour market conditions (Bozick, 2009 for the USA; Sievertsen, 2016 for Denmark).

Therefore, contrary to the *economic resources hypothesis*, an *opportunity costs hypothesis (H2)* expects that rising unemployment rates increase postsecondary enrolment among youth whose parents do not have tertiary education but do not affect the enrolment of those from tertiary-educated families.

Figure 1 summarizes hypotheses 1 and 2 by illustrating how enrolment changes with the increasing unemployment rate. The *economic resources hypothesis* (H1) expects that the slope is the steepest for students from tertiary-educated families, followed by students whose parents have middle level of education (upper secondary), while the enrolment likelihood rather decreases for students whose parents have a low level of education. In contrast, the *opportunity costs hypothesis* (H2) expects that the slope is steeper for students from low-educated and middle-educated families whose decisions might depend more on opportunity costs as they might achieve a social status similar to their parents also without enrolling to university.

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2.4 Affordability of tertiary education

Institutional contexts might moderate the extent to which economic resources and uncertainties as well as opportunity costs affect enrolment decisions during an economic downturn. Although many aspects of institutional context likely contribute to such moderation, we here focus on the affordability of postsecondary education and discuss whether it might promote the enrolment opportunities of youth from lower social backgrounds during recessions in particular. Overall, tertiary education systems tend to be more egalitarian if disadvantaged social groups are strongly supported or if the funding comes largely from public resources (Marginson, 2016). Empirical research suggests that lowering the costs of higher education particularly by need-based grants or subsidized tuition can increase enrolment and improve academic outcomes for students from less advantaged backgrounds (Dynarski and Scott-Clayton, 2013; Goldrick-Rab *et al.*, 2016). Compared across countries, the degree of affordability of tertiary education differs considerably: the Nordic countries combine free tuition with government loans or grants (see the overview in Thomsen *et al.*, 2017), while Anglo-Saxon countries balance the costs of tuition with student loans and grants (e.g. the typology in Triventi, 2014).

Over and above the actual costs of education, children and their families make educational decisions based on their subjective evaluation of costs and benefits (Breen and Goldthorpe 1997). These perceptions can differ by social class due to available information or class-specific constraints and preferences for status maintenance (Grodsky and Jones 2007; Abbiati *et al.*, 2018). Such differences in the evaluation of costs and benefits might play a more important role in institutional contexts where the enrolment decision involves significant financial risk. For instance, previous research has demonstrated that students from less advantaged backgrounds have stronger debt-averse attitudes than their peers from privileged backgrounds (Callender and Mason 2017) and are also more vulnerable to the risks associated with student loan debt (Houle 2014).

As the affordability of tertiary education likely shapes the perceptions of opportunity costs and the relevance of economic resources and uncertainties, we propose an *affordability hypothesis* (H3). This hypothesis posits that the affordability of tertiary education enhances the chances of students from lower social backgrounds to enrol in university in response to an increase in unemployment rates. Thus, if a recession increases social inequalities, as expected in the economic resources hypothesis (H1), affordability is likely to moderate this recession-related increase in enrolment gaps. If a recession reduces inequalities, as expected in the opportunity costs hypothesis (H2), the affordability of tertiary education is likely to further enhance this recession-related reduction of inequalities.

3. Data and variables

3.1 Harmonized dataset and transition outcomes

Our empirical analysis is based on data from five longitudinal surveys covering educational transitions in the years 2004–2016. The main source of data is the EU-SILC longitudinal files (2007–2017) from where we obtained data for 28 countries (see Table 1 for the list of countries). In addition, we obtained British data from the British Household Panel Survey (2003–2008) and its successor study Understanding Society (2009–2016). The data for Germany is from the German Socio-Economic Panel (2002–2017). For the USA, we used the SIPP panels 2004 and 2008, which cover educational transitions occurring between

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Table 1. Contextual variables, over years 2004–2016

	Unemployment rate[a]				Affordability of tertiary education[b]
	Mean	Min	Max	Range	Value in 2005
Austria	5.2	4.1	6.0	1.9	16
Belgium	8.0	7.0	8.5	1.5	14
Bulgaria	9.7	5.6	13.0	7.4	6
Croatia	13.2	8.6	17.4	8.8	3
Cyprus	8.7	3.7	16.1	12.4	28
Czech Republic	6.4	4.0	8.3	4.3	5
Denmark	6.0	3.7	7.8	4.1	30
Estonia	8.4	4.6	16.7	12.1	6
Finland	8.1	6.4	9.4	3.0	16
France	9.3	7.4	10.4	3.0	7
Germany	7.1	4.1	11.2	7.1	17
Greece	16.4	7.8	27.5	19.7	1
Hungary	8.4	5.1	11.2	6.1	12
Ireland	9.9	4.6	15.5	10.9	12
Iceland	4.6	2.3	7.6	5.3	21
Italy	9.2	6.1	12.7	6.6	12
Latvia	11.8	6.1	19.5	13.4	5
Lithuania	10.4	4.3	17.8	13.5	11
Luxembourg	5.2	4.2	6.5	2.3	10
Malta	6.3	4.7	7.2	2.5	1
Netherlands	5.6	3.7	7.4	3.7	19
Norway	3.7	2.5	4.8	2.3	41
Poland	10.6	6.2	19.1	12.9	1
Portugal	11.5	7.8	16.4	8.6	6
Romania	6.8	5.6	8.0	2.4	4
Slovakia	13.2	9.6	18.4	8.8	11
Slovenia	7.3	4.4	10.1	5.7	18
Spain	17.3	8.2	26.1	17.9	6
Sweden	7.5	6.1	8.6	2.5	24
UK	6.2	4.7	8.1	3.4	15
USA	6.6	4.6	9.6	5.0	10
Total	8.7	2.3	27.5	25.2	12

Sources:
[a]Eurostat 2019a.
[b]OECD (2018), World Bank (2016), and Eurostat (2019b).

2004 and 2013. All these surveys provide nationally representative household samples (see Supplementary Appendix 1 for more details). Although the EU-SILC includes Germany and the UK, we decided to use national longitudinal studies to overcome some data limitations in the EU-SILC (see below).

We analyse data for young people who within the timeframe of the surveys completed upper secondary education (see also footnote 2 in Supplementary Appendix 1). For this, we harmonized data in all five surveys to create a new youth dataset. The new harmonized dataset

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includes 36,352 individuals who are nested in 370 country-transition years within 31 countries (see Supplementary Appendix 1 for detailed information about the harmonization of data and variables). We use the term transition year for the year that a student completed upper secondary education. It is important to note that we excluded a very small number of secondary school graduates (<1 % of the sample) who did not reside in the same household with at least one of their parents before finishing school.

We measure the transition outcomes of young people about 1 year after completing upper secondary education. Three binary dependent variables indicate transition outcomes: (a) enrolment in postsecondary education versus all other outcomes; (b) unemployment and inactivity that is not related to participation in education (NEET) versus all other outcomes; (c) enrolment in postsecondary education versus entry to employment. Although young people might have double statuses (e.g. working students), our categorization mainly relies on the respondents' own perception of their main activity (see Supplementary Appendix 1 for further details). In total, about 65% of young people continued studies in postsecondary education, 21% entered the labour market and 14% entered NEET status. Note that our measurement of enrolment involves all types of postsecondary education, such as applied and academic tertiary education as well as non-tertiary postsecondary education. Unfortunately, the EU-SILC longitudinal files do not allow any further differentiation. Thus, our results reflect an overall propensity to invest in postsecondary education. Moreover, apprenticeships and other types of in-firm training positions are coded under employment due to the EU-SILC coding (see Supplementary Appendix 1).

Our sample for Germany, the UK and the USA included only school-leavers eligible for postsecondary education (see additional details in Supplementary Appendix 1). However, the EU-SILC longitudinal files do not contain information regarding the type of upper secondary education. While we acknowledge this as an important limitation for our analysis, our focus on overall entry to postsecondary education should reduce potential biases. Several countries offer non-tertiary postsecondary education options accessible to graduates of vocational secondary schools. Further, to address the possible variance introduced by the type of secondary education, our analysis includes a time-varying control variable for the vocational orientation of secondary education systems and its interactions with social background measures (see Section 4).

3.2 Data for regions

While the natural focus in comparative cross-national research tends to be on the national-level context, we will additionally employ regional unemployment rates in order to maximize leverage and empirical variation in our contextual analysis. Although research focusing on single countries has shown that employment opportunities and occupational structure of the local labour markets affect students' enrolment decisions to postsecondary education (e.g. Bozick 2009; Ananat *et al.*, 2017), less is known about heterogeneous effects of regional unemployment in the comparative context.

In this, our analysis relies on data for states in the USA and regions in 13 larger European countries with data for region identifiers: Austria, Belgium, Bulgaria, Germany, France, Greece, Hungary, Italy, Poland, Romania, Spain, Sweden and the UK. We focus on NUTS 1 level regions (Eurostat 2020), which are major socio-economic regions within countries (e.g. federal states in Germany). We excluded smaller countries from this analysis as they are classified as only one NUTS 1 region by Eurostat, that is, their national and regional levels overlap. The sample enriched with corresponding regional data include 18,827

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respondents from 1,337 regions. Unfortunately, we could not include the Netherlands, Portugal and Norway in this analysis as the EU-SILC does not provide region identifiers for these three countries. Although the choice of geographical unit was determined by data availability, often decisions about educational policies are taken at the level of state or larger region and regional unemployment rates are likely a reasonable proxy of job opportunities available to young people in their larger commuting area.

3.3 Individual-level variables

Our main interest is how macroeconomic conditions interact with social background, which we measure with parental education. We chose to use parental education instead of social class as an indicator of social background because this variable is more standardized across countries and because the number of missing values is smaller, especially during a recession. Additionally, some research suggests that parental education has a more significant impact on children's educational attainment than social class (e.g. Bukodi and Goldthorpe, 2013). In our analysis, we considered both maternal and paternal educational levels and identified the *highest level of parental education* that contains four categories: (a) tertiary education (we refer to them as tertiary-educated families), (b) postsecondary non-tertiary, (c) upper secondary (i.e. middle-educated families) and (d) lower secondary or less (i.e. low-educated families). As only 7% of parents had postsecondary non-tertiary education in our sample (Table 2), we do not specifically focus on interpreting the results for this group in our analysis. Moreover, we had to exclude a small number of respondents (about 1% of the sample) who did not have any information about parental education.

Table 2. Distribution of individual level variables, sample divided by transition outcomes

	Studies	Employment	NEET	Total
Female	0.54	0.42	0.45	0.50
Age of completing upper secondary education	18.9	19.6	19.3	19.1
SD	(1.0)	(1.5)	(1.5)	(1.3)
Military spell	0.01	0.02	0.02	0.01
Parental education				
Tertiary	0.42	0.28	0.23	0.37
Postsecondary non-tertiary	0.07	0.08	0.07	0.07
Upper secondary	0.42	0.49	0.50	0.45
Lower secondary or less	0.09	0.15	0.20	0.11
Single-parent household	0.17	0.19	0.24	0.19
Number of children aged <16 years in the household				
0	0.59	0.58	0.59	0.59
1	0.29	0.27	0.28	0.28
2	0.09	0.11	0.10	0.09
3 or more	0.03	0.05	0.04	0.04
Parent was unemployed	0.11	0.12	0.20	0.13
Income decile	5.7	5.2	4.6	5.4
SD	(2.9)	(2.7)	(2.7)	(2.8)
N	23,461	7,843	5,048	36,352

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We also analyse whether household income and parental unemployment explain why the effect of parental education varies with macroeconomic conditions. The variable *parental unemployment* indicates that at least one parent in the household was recently unemployed (in the transition year or a year before). Further, we calculated *family income deciles* by years and countries based on household equalized disposable income after tax and other deductions. We used the modified equivalence scale by the Organisation for Economic Co-operation and Development (OECD) to adjust the income to household composition. Household income is measured for the year before the respondent completed upper secondary education and was still living in the parental household.

To account for demographic variations and differences in the composition of school-leavers across countries, our analysis includes control variables for *gender*, living in a *single-parent household*, the *number of children* aged less than 16 in the household, *age* of completing upper secondary education and a spell of *military service*. If a student entered compulsory military service after leaving school (only in a few countries in our sample), we treated the status a year after military service as the status after leaving school (see Supplementary Appendix 1 for more information about the definition of variables and missing values).

3.4 Unemployment rate

The annual unemployment rate is the main explanatory variable that describes the macroeconomic context (data are from Eurostat 2019a; The United States Department of Labor 2014). The *unemployment rate* denotes the percentage of unemployed individuals from the total workforce (between ages 15–74 years). The definition of unemployed aligns with the guidelines of the International Labour Office and includes those without work, currently available for work and seeking work as well as those without work who had found a job to start within a period of at most 3 months. We also tested our models with measures of youth unemployment and with the unemployment rate among prime-age workers and found very similar trends, which is not surprising as different measures of unemployment rates are strongly correlated (see Supplementary Appendix 2). We opted against using the youth unemployment rate due to concerns of endogeneity, given that the youth unemployment rate is inherently tied to our dependent variable, that is, experiencing unemployment is a part of the transition outcomes we are examining.

We used the unemployment rate in the year a student completed upper secondary education. In additional tests, we also explored the importance of the change in the unemployment rate compared to earlier years.

Table 1 shows that unemployment rates vary greatly over the 2004–2016 period and across countries in our sample. On average, unemployment rates were the lowest in Norway, Iceland, Austria, Luxembourg and the Netherlands. We observe the highest unemployment rates in some Southern and Eastern European countries. Specifically, unemployment rates peaked in Greece, Spain, Croatia and Portugal in 2013, in the Baltic countries in 2010 and in Poland and Slovakia in 2004.

3.5 Other contextual variables

We created an index of the affordability of tertiary education by multiplying a share of *financial support to students* with a share of *public funding* in tertiary education, and then dividing the product by 100. First, *financial support* is measured as a percentage of financial

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aid to students (e.g. transfers, social benefits and loans) from the total public expenditure on education at the tertiary level of education (data are from Eurostat (2019b), see also Eurostat (2009): 207). Second, *public funding* indicates a percentage of public expenditure on tertiary education institutions from the total expenditure on tertiary education, that is, the sum of public and private expenditure (data are from OECD (2018), except for Bulgaria, Croatia, Cyprus, Romania and Malta from World Bank (2016)). We use data for the year 2005 for both measures, as the data for public funding are not available for 2004 (i.e. the beginning of our observation window). By using a time-invariant indicator for affordability, we aim to exclude the possible endogenous effects of policy responses to increasing unemployment rates from our analysis. Nevertheless, we conducted additional tests to explore the significance of a time-varying variable for the affordability of tertiary education. Table 1 shows that in 2005, Norway, Denmark, Cyprus, Sweden and Iceland had the highest affordability index, while Greece, Malta, Poland and Croatia had the lowest, mainly attributed to minimal financial support for students.

Finally, our models include a time-varying control measure for the *vocational orientation* of the secondary education system defined as a percentage of pupils enrolled in vocational studies at the level of upper secondary education (data from Eurostat, 2019b). This measure does not include postsecondary vocational studies. Besides between-country differences, this measure controls for the change of vocational orientation over time. Including vocational orientation in our models serves two purposes. First, it reduces the possible effects of differences in secondary education systems, which could impact how the unemployment rate affects transition probabilities to postsecondary education. Second, controlling for vocational orientation of the secondary education helps to reduce system-specific variance between countries in the risk of NEET entry, as stronger vocational orientation tends to reduce information asymmetries between education and the labour market and thus, facilitate youth entry to the first job (e.g. comparative analysis in Barbieri *et al.*, 2016).

4. Method

We aim to estimate the effect of aggregate unemployment rates on our three dependent variables that measure the transition outcomes of secondary school graduates. Instead of using logistic models, we employ linear probability models because they enable comparisons of coefficients across the same-sample nested models (see also Karlson *et al.*, 2012).

Our model starts with considering a three-level structure of data: graduates (i) are nested in country–years (j) and these country–years are nested within countries (c). At the country level, we use country fixed-effects estimator (FE_c) to control for unobserved heterogeneity across countries. Thus, we rely on within-country variance in unemployment rates in estimating their effect on transition outcomes. This helps to reduce possible bias due to systematic differences between countries in their educational systems and labour markets. The resulting basic model can be written as follows:

$$Pr(Y_{ijc}) = \beta_0 + \beta_1 SES_{ijc} + \beta_2 X_{ijc} + \beta_3 UE_{jc} + \beta_4 Z_{jc} + FE_c + u_{jc} + e_{ijc}, \qquad (1)$$

where the predicted probability of transition outcome (Y_{ijc}) depends on parental education ($\beta_1 SES_{ijc}$) and individual-level control variables ($\beta_2 X_{ijc}$) as well as on the unemployment

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rate ($\beta_3 UE_{jc}$), and the country–year-level control variable for vocational orientation ($\beta_4 Z_{jc}$). In addition to country dummies (FE_c), the model includes country–year-level random intercept (u_{jc}). e_{ijc} denotes random variation at the individual level.

To explore how unemployment affects inequality in postsecondary education outcomes, we estimate cross-level interactions between parental education and the unemployment rate by including the interaction term $\beta_5(SES_{ijc} * UE_{jc})$ and random slopes for parental education in Equation (1). These models also control for the interaction between parental education and vocational orientation of secondary education.

We also explore to what extent changing household resources during recessions explain changing social inequality in transition outcomes. For this, we add to our models interaction terms (a) between household income and unemployment rate and (b) between parental unemployment and unemployment rate. These models include random slopes for income and parental unemployment. Moreover, we investigate whether the affordability of tertiary education mitigates the extent to which unemployment rates affect social inequality. For this, we include cross-level interactions between parental education and affordability into our models.

The additional analysis of regional unemployment rates considers a four-level structure of the data: graduates are nested in region–years, which are nested within regions and regions within countries. Hence, our models for regional effects include in addition to fixed effects for countries also intercepts for regions and region–years. As regional unemployment rate is a region–year level variable, the models testing cross-level interactions with parental education also include slopes for parental education at this level.

5. Results

5.1 The effect of unemployment rate

We begin with models that estimate an average effect of the unemployment rate on transition outcomes. Figure 2 presents the results of multilevel regression analysis and Supplementary Appendix 3 shows the corresponding table of estimates. These models include fixed effects for countries that reduce possible bias due to systematic differences between countries.

The findings show that higher unemployment rates somewhat increase the likelihood for entry to postsecondary education compared to all other possible transition outcomes (first graph in Fig. 2). The model with fixed effects shows that a 1 percentage point increase in unemployment rate increases the predicted probability to attend postsecondary education by about 0.7 percentage points. In addition, we also find a small positive effect of the unemployment rate on the risk of entry to unemployment or inactivity, that is, NEET status (second graph in Figure 2). Finally, a model that compared continuing studies with entry to employment, which is a comparison excluding adolescents who entered NEET status, shows that a 1 percentage point increase in the unemployment rate increases the likelihood for entry to postsecondary education instead of getting a job by about 0.9 percentage points (third graph in Figure 2).

5.2 Heterogeneity by social background

We next turn to our main question on whether social inequality in transition outcomes is increasing during a recession. We begin to explore how the effect of the unemployment rate

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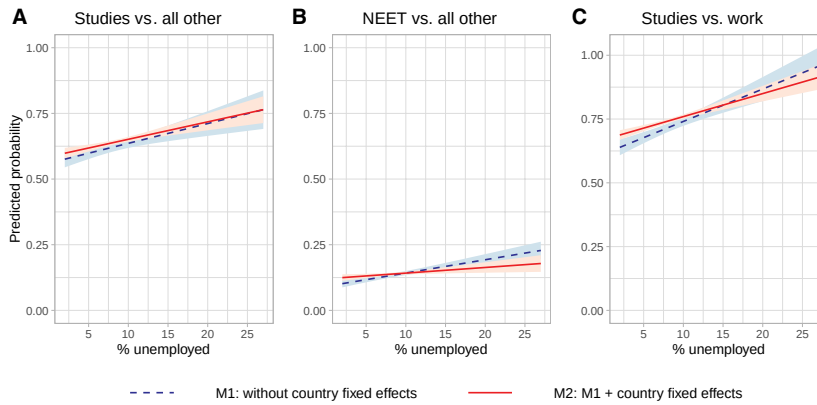

Figure 2. The effect of unemployment rate on transition outcomes.

Notes: Predicted probabilities with 95% confidence intervals from multilevel linear probability models. Models control for gender, military spell, parental education, single-parent household, number of children in household, age of completing upper secondary education and country–year level variable for vocational orientation of secondary education. See Supplementary Appendix 3 for full models.

depends on social background by estimating interactions between parental education and unemployment rate (Model 1a–c in Table 3 and Figure 3). Thereafter, we test whether lower household resources during economic recessions might explain the increasing inequalities in transition outcomes. For this, we add into our models the interactions between unemployment rate and household resources, which are measured with family income decile and parental unemployment (Model 2a–c in Table 3). Finally, we test the role of opportunity structure by including in the models the interactions between parental education and affordability of tertiary education (Model 3a–c in Table 3 and Figure 3).

As a first step, we compare entry to postsecondary education with all other possible transition outcomes. The findings show that when the unemployment rate is higher, then the likelihood for enrolment increases for young people whose parents have tertiary or upper secondary education while youth from low-educated households do not benefit from increasing levels of unemployment (Model 1a in the first column of Figure 3 and Table 3). More specifically, predicted probabilities in Figure 3 indicate that a 1 percentage point increase in unemployment rate increases the likelihood of enrolment for youth from tertiary-educated families by 0.8 percentage points and from middle-educated families by 0.6 percentage points but there is almost no change for their peers from low-educated households. Hence, adverse macroeconomic conditions somewhat increase social inequalities in youth transitions, as young people whose parents have low education stay behind. However, the overall magnitude of this difference is not large. For example, if unemployment rate increases by 10 percentage points, then the entry gap between youth from tertiary-educated and low-educated families increases by 7 percentage points.

Overall, these findings, indicating an increase in inequality with rising unemployment rates, align with the expectations of the *economic resource hypothesis* (H1). However, contrary to this hypothesis, we do not observe a reduction in enrolment for youth from

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Table 3. Interacting effects of national-level unemployment rate and parental education on transition outcomes

	(A) Studies versus all other			(B) NEET versus all other			(C) Studies versus work		
	Model 1a	Model 2a	Model 3a	Model 1b	Model 2b	Model 3b	Model 1c	Model 2c	Model 3c
Unemployment (UE) rate	0.008***	0.005*	0.003	−0.002*	0.001	0.002	0.007***	0.006**	0.005*
	(0.002)	(0.002)	(0.002)	(0.001)	(0.002)	(0.002)	(0.001)	(0.002)	(0.002)
UE rate * parental education (ref. tertiary)									
Postsecondary non-tertiary	0.001	0.003	0.005+	0.006**	0.004*	0.002	0.005+	0.006**	0.008**
	(0.002)	(0.003)	(0.003)	(0.002)	(0.002)	(0.002)	(0.003)	(0.002)	(0.003)
Upper secondary	−0.002	0.001	0.004**	0.005***	0.003*	0.000	0.002+	0.003*	0.005**
	(0.002)	(0.002)	(0.002)	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)	(0.002)
Lower secondary or less	−0.007**	−0.004	−0.000	0.012***	0.008***	0.007***	−0.000	0.001	0.004
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.003)
UE rate * parent unemployed		−0.005**	−0.005**		0.003+	0.003*		−0.003+	−0.003*
		(0.002)	(0.002)		(0.001)	(0.001)		(0.002)	(0.002)
UE rate * income decile		0.001**	0.001**		−0.001**	−0.001**		0.000	0.000
		(0.000)	(0.000)		(0.000)	(0.000)		(0.000)	(0.000)
Affordability * parental education									
Postsecondary non-tertiary			0.003+			−0.002+			0.002
			(0.002)			(0.001)			(0.001)
Upper secondary			0.004***			−0.003***			0.002**
			(0.001)			(0.001)			(0.001)
Lower secondary or less			0.005**			−0.003*			0.003*
			(0.001)			(0.001)			(0.001)
Slopes for parental education	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Slopes for income and parent unemployed	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Observations		36,352			36,352			31,304	
Number of groups		370			370			369	

Notes: Coefficients from multilevel linear probability models with country fixed effects. Models include main effects for all interacted variables (see Supplementary Appendix 11), control variables at individual level (gender, military spell, single-parent household, number of children in household, and age of completing upper secondary education), vocational orientation and its interactions with parental education, and country–year-level intercept. Standard errors in parentheses.

Statistical significance levels at

$^{+}P < 0.1$.

$^{*}P < 0.05$.

$^{**}P < 0.01$.

$^{***}P < 0.001$ based on two-tailed tests.

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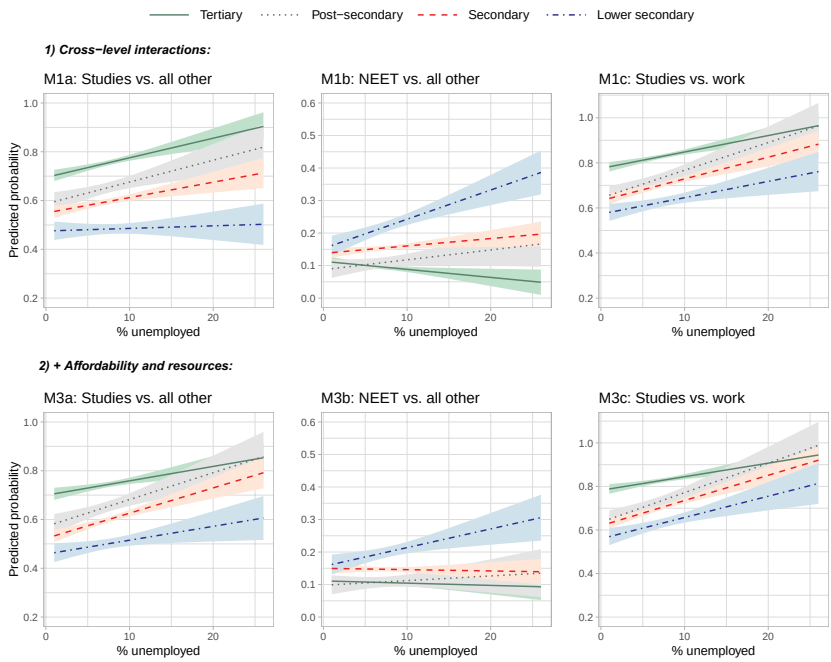

Figure 3. Interacting effects of national-level unemployment rate and parental education.
Notes: Predicted probabilities with 95% confidence intervals from multilevel linear probability models. The same models are presented in Table 3.

lower-educated families. On the other hand, the observed pattern does not align with the expectations of the *opportunity costs hypothesis* (H2).

Next, we control for household income and parental unemployment, as well as for their interaction with unemployment rate (Model 2a in Table 3). After including these variables, the entry gap between youth from tertiary-educated and low-educated families reduces and is not statistically significant anymore. This finding points towards a significant role of family income and employment opportunities in explaining the larger entry gap during recessions and is consistent with the *economic resources hypothesis* (H1) expecting inequalities to increase during recession. The model also shows that youth from households with lower income and unemployed parents are less likely to continue their studies when unemployment rates are higher (see Supplementary Appendix 4 for additional details).

Finally, we include a control for the time-invariant measure for the affordability of tertiary education (Model 3a in Figure 1 and Table 3), which accounts for differences in education policies across countries and may partly also reflect their welfare systems (see Section 4). We find that if tertiary education were similarly affordable for all social groups, then the unemployment rate would increase the likelihood of enrolment slightly more for youth from middle-educated families compared to their peers from tertiary-educated families (the difference becomes statistically significant in Model 3a as opposed to earlier models). As an additional robustness check, we tested interactions between parental education, unemployment and affordability (Supplementary Appendix 5). However, these three-way interactions

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did not reach statistical significance, possibly due to the small sample size at the macro level, and thus, caution against strong conclusions. In contrast, we do not find similar results for youth from low-educated families, although there is some indication that the larger enrolment gap between them and youth from tertiary-educated families during recessions reduces after accounting for affordability. Overall, the findings offer only limited support for the *affordability hypothesis* (H3), which expected that more affordable contexts encourage young people from less advantaged families to enrol in postsecondary education in response to increasing unemployment rates.

As a second step in our analysis, we explore how the unemployment rate affects the extent of social inequality in the risk of NEET entry. The findings in Model 1b show that a higher unemployment rate is associated with a higher NEET risk for adolescents from low-educated households: a 1 percentage point increase in unemployment rate rises their NEET entry risk by 0.9 percentage points (Figure 3 and Table 3). In contrast, increasing unemployment rate does not affect the NEET risk for youth from tertiary-educated families (a 1 percentage point increase by −0.2 percentage points). Model 2b in Table 3 shows that the gap between these two groups reduces after controlling for family economic resources but remains statistically significant. It also indicates that parental unemployment and lower family income increase NEET risks during recessions. Furthermore, we find that young people whose parents have upper secondary education are somewhat more likely to enter NEET when the unemployment rate rises compared to youth whose parents have tertiary education. This difference decreases after controlling for economic resources and affordability (note that the three-way interactions between parental education, unemployment and affordability were not statistically significant). Overall, our findings align with the *economic resources hypothesis* (H1), suggesting that inequalities in the risk of NEET entry increase with rising unemployment rates. However, we cannot draw strong conclusions here due to the moderate importance of the moderating role of family economic resources in our empirical analysis.

As a third step of analysis, we explore transition outcomes in further detail by contrasting enrolment to postsecondary education with entry to employment, a comparison that excludes the risk for NEET. The findings provide some indication that a rising unemployment rate increases the likelihood of choosing postsecondary education over employment slightly more for young people from middle-educated families than for their peers from tertiary-educated families (Model 1c in the second column of Figure 3 and Table 3). It is important to note that we did not find such differences in the overall enrolment rate but only when we exclude labour market risks, which, under adverse macroeconomic conditions, are higher for youth from middle-educated families than for youth from advantaged backgrounds (Models 1a and b). Models 2c and 3c reveal that, in the case of similar parental resources and affordability, the unemployment rate further increases the likelihood of youth from middle-educated families enrolling compared to their tertiary-educated counterparts. However, once again, the three-way interactions between parental education, unemployment and affordability did not reach statistical significance (Supplementary Appendix 5). Overall, the observed pattern suggests that opportunity costs might be a relevant consideration for young people from middle-educated families as their likelihood to enrol increases with unemployment rate. Hence, these findings provide some limited support for the *opportunity costs hypothesis (H2)*, expecting that opportunity costs of studies in terms of forgone

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Table 4. The effect of regional unemployment rate.

	(A) Studies versus all		(C)[AQ] NEET versus all		(B) Studies versus work	
	M1a	M2a	M1b	M2b	M1c	M2c
Regional unemployment rate	0.005***	0.007***	0.003**	−0.001	0.008***	0.007***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Regional unemployment rate * parental education (ref. tertiary)						
Postsecondary non-tertiary		0.001		0.003		0.004
		(0.003)		(0.002)		(0.003)
Upper secondary		−0.000		0.004**		0.004*
		(0.002)		(0.001)		(0.002)
Lower secondary		−0.003+		0.007***		0.002
		(0.002)		(0.002)		(0.002)
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Intercept for region	Yes	Yes	Yes	Yes	Yes	Yes
Intercept for region–year	Yes	Yes	Yes	Yes	Yes	Yes
Slope for parental education (region–year level)	No	Yes	No	Yes	No	Yes
Observations	18,827		18,827		16,049	
Number of groups	1,337		1,337		1,302	

Notes: Coefficients from multilevel linear probability models with country fixed effects. Models include main effects for all interacted variables, control variables at individual level (gender, military spell, single-parent household, number of children in household, and age of completing upper secondary education), vocational orientation and its interactions with parental education, and intercepts for region and for region–year level. Standard errors are in parentheses.
Statistical significance levels at
$^{+}P < .1$.
$^{*}P < .05$.
$^{**}P < .01$.
$^{***}P < .001$ based on two-tailed tests.

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labour market opportunities might increase enrolment rates for students from lower social backgrounds during recessions.

As a last step of analysis, we investigate the role of regional unemployment rate in affecting inequality in access to postsecondary education. Models 1a–c in Table 4 show that similarly with the national unemployment rate, a higher regional unemployment rate increases the likelihood that young people enter postsecondary education. Models 2a–c present the interaction effects between parental education and regional unemployment rates, which resemble our findings for the national unemployment rate. Importantly, we find that compared to their peers from tertiary-educated families, young people from low-educated families are slightly less likely to enter postsecondary education when regional unemployment rates are high, while their risk of NEET entry increases. Moreover, higher regional unemployment rates increase the likelihood for enrolment for both youth from middle-educated and tertiary-educated families, and especially for those from middle-educated families when different risks of NEET entry are considered.

5.3 Robustness checks

On the institutional side of our analysis, it might be a concern that our preferred measure for the affordability of tertiary education relies on between-country variance only. However, the funding of postsecondary education might decline in response to a recession, which could lead to higher tuition costs for students. For instance, in the USA, the increase in enrolments and reductions in state budget allocations during the period of the Great Recession accelerated the shifting of costs of postsecondary education from states to students, despite the increase in student grants (Barr and Turner 2013; Dynarski and Scott-Clayton 2013). To address this concern, we estimated additional models that included annual measures for the change in affordability relative to 2005, alongside the baseline time-constant measure of affordability in the year 2005 (see Supplementary Appendix 6). We find that including changes in affordability to our models does not affect the interaction terms between parental education and the unemployment rate, but also that changes in the affordability of tertiary education do not seem to affect inequality of access to postsecondary education in the first place. It is important to note, however, that this latter result is probably more indicative of the limited nature of institutional changes in affordability over just a decade of data, and thus also corroborates our methodological decision to base our main analysis on an indicator of durable institutional differences between national systems of postsecondary education.

With respect to the latter, an additional consideration arises with regard to whether our affordability index may sufficiently account for the substantial differences in tuition rates between the USA and European countries, despite indirectly capturing these differences as part of private expenditure on tertiary education. Unfortunately, to the best of our knowledge, there is no alternative affordability index available for all the countries in our sample. To explore the potential impact of the USA being an outlier due to its high tuition fees, we estimated the same models without including US data and the results were similar (see Supplementary Appendix 7).

We acknowledge that despite fixed effects for countries, cross-level interactions are not independent of between-country variation (Giesselmann and Schmidt-Catran 2019). Hence, as a robustness check, we tested models that use within-country variance in estimating the effect of unemployment rate on the extent of social inequality. For this, we controlled for country effect heterogeneity in unemployment rate and in parental education by interacting country dummies with unemployment rate and with parental education

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(Supplementary Appendix 8). For example, bias might arise due to varying labour market policies across countries that could impact the unemployment rate, as well as how vulnerable youth from lower-educated families are to the NEET risk. Consistent with our main models in Table 3 (Model 1a–c), we find that even in these very restricted models the interaction effects between parental education and unemployment rate have similar direction albeit somewhat smaller effect sizes.

Besides the impact of current macroeconomic conditions at the transition year, the magnitude of recent changes in these conditions might affect youth transitions. Thus, we tested models that include both the unemployment rate and its change compared to the previous year (Supplementary Appendix 9). We find that when the yearly unemployment rate is controlled for, the recent increase in the unemployment rate slightly increases the overall risk of NEET entry and the likelihood that young people continue their studies in postsecondary education instead of entering into employment.

To account for the potential influence of the overall secular trend of educational expansion across all countries overtime on the results, we included year fixed effects in the models estimating the probability of enrolment. Supplementary Appendix 10 shows that the results remained consistent in these more complex models.

Finally, we tested models where we included slopes for all individual level variables (not presented) and the results were identical to Table 3.

6. Conclusion

Recent economic challenges have raised concerns regarding their impact on the inequalities in the life courses of young people. Although extensive research suggests that economic downturns increase postsecondary enrolment of young people (e.g. Clark, 2011; Barr and Turner, 2013), their implications on the inequality of educational opportunity are less clear. This article examined how macroeconomic conditions affect the extent of social inequality in postsecondary transitions. Using longitudinal data from 30 European countries and the USA, we analysed the effect of the unemployment rate on youth transitions over a period including the Great Recession.

Empirically, we find that higher unemployment rate increases postsecondary enrolment, which is a consistent finding with prior research in economics (e.g. Clark, 2011; Barr and Turner, 2013). However, we also find that this impact is not even across social backgrounds. In accordance with the economic resources hypothesis, adverse macroeconomic conditions slightly increase social inequalities in entry to postsecondary education, as the gap in enrolment rates increases between adolescents from tertiary-educated and low-educated families. Yet, the magnitude of this effect is relatively small. This could be seen also as a positive indication that educational systems to some extent might manage to contain the impact of recessions on inequality in access to postsecondary education. However, we also find that the risk of poor transition outcomes, such as unemployment and inactivity, increases especially for young people from low-educated families during recessions. The analysis suggests that changing income and employment insecurity during recessions are relevant explanations for slightly increasing inequality in postsecondary transitions. Hence, as economic recessions disproportionally affect less advantaged families (Dellas and Koubi, 2003), it seems more difficult for them to invest in their children's education or to provide the necessary support for a smooth entry to the labour market (e.g. through social networks). This finding aligns with a comparative study by Salazar *et al.*

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(2020) demonstrating that a recession can heighten the degree of inequality in educational expectations. Our findings further indicate that this shift in expectations could be accompanied by disparities in transition outcomes.

The finding that rising unemployment rates slightly increase the likelihood of postsecondary enrolment for youth from tertiary-educated families suggests the relevance of the economic resource perspective, as advantaged families typically have more resources to provide extensive safety nets (Mayer, 2001) to mitigate the negative consequences of recessions on postsecondary transitions. It is important to note that enrolment rates are high for youth from advantaged families also during favourable economic conditions, which indicates a strong aspiration for status maintenance within this group. However, it is plausible that their aspiration for status maintenance interacts with uncertainties related to adverse macroeconomic conditions, where falling behind might be perceived as risky in terms of maintaining social status. For instance, some previous research has suggested that class inequality in enrolment might increase during periods of heightened competition, as advantaged families adapt by making considerable efforts to secure the intergenerational transmission of social status (Alon, 2009). Furthermore, they might respond by investing in more prestigious universities. However, given the data limitations, we have to leave it to future research to explore how economic downturns affect the relationship between social inequalities and differentiation within the tertiary education system (Lucas, 2001).

We find that rising unemployment rates increase the likelihood of postsecondary enrolment also for young people from middle-educated families (i.e. parents with upper-secondary education). If their parents' incomes and employment are less affected, and tertiary education is more affordable, they are even more likely to enrol in response to increasing unemployment rates than their counterparts from tertiary-educated families. However, their vulnerability to the risk of NEET entry also somewhat increases during recessions. As adolescents from middle-educated families would avoid downward educational mobility even without attending tertiary education, possibly opportunity costs are a relevant consideration for them, that is, a choice between immediate income from the labour market and long-term investment in education (Dellas and Sakellaris, 2003; Bozick, 2009). In addition, the financial situation of middle-educated families is likely to be somewhat better during recessions compared to low-educated families, for whom lacking economic resources might outweigh the considerations of opportunity costs. These two groups might also differ in aspirations, as previous research on higher education expansion suggests that the increase in enrolment relates to the high aspirations of expanding middle classes (Marginson, 2016).

Furthermore, our findings for young people from middle-educated families tentatively suggest that, during adverse macroeconomic conditions, more affordable tertiary education may increase their likelihood of enrolling in education. In general, the affordability of postsecondary education can lower the actual and perceived costs of postsecondary education (Dynarski and Scott-Clayton, 2013) and thus, promote the enrolment of less advantaged students during recessions. It is important to note that the affordability of tertiary education tends to correlate with the generosity of the welfare state across countries (Triventi, 2014). While some research has suggested that social class influences on a child's educational outcomes are not easily mitigated by welfare state programmes (Heckman and Landersø, 2022), previous evidence focusing on the dynamic aspects of stratification has shown that more generous unemployment benefits could improve access to postsecondary education for unemployed families (Lindemann and Gangl, 2020). Hence, a part of the impact

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associated with affordability in our analysis may reflect a higher perceived financial security among less advantaged families during periods of high unemployment due to more generous welfare state support rather than higher education policies per se. An important limitation, however, is that our affordability index does not fully account for the fact that the USA generally has much higher tuition rates than European countries, although it indirectly captures these differences as part of private expenditure on tertiary education.

In this study, we used country fixed effects along with extensive robustness checks to limit the possible confounding effects of unobservable characteristics on cross-level interactions of interest. However, the lack of comparative cross-national data that would allow us to estimate family fixed-effects models to more effectively control for additional unobserved family-level factors restricts us from making stronger causal claims in this article. Furthermore, although our results suggest that financial resources are relevant in explaining the effect of economic downturns on social inequality in youth transitions, we were not able to empirically test the role of other mechanisms, such as educational aspirations, school achievement or the psychological effects of recessions on families (cf Kalil, 2013). Moreover, our focus on inequality among young people who have successfully completed upper secondary education may underestimate the overall impact of a recession on educational inequality, as it could disproportionally increase school dropout rates among youth from low-SES families, whose parents are more likely to experience unemployment. Therefore, addressing this question is crucial for future research.

Nonetheless, this study highlights that economic recessions induce a risk for postsecondary transitions of youth from low-educated families. A targeted policy approach may be essential to address the challenges faced by young people from lower social classes during recessions, such as targeted financial support, labour market programmes, or career guidance. In contrast, our findings indicate that high enrolment rates of youth from advantaged families even slightly increase in the face of adverse economic conditions. Yet, increasing unemployment rates seem to incentivize also their counterparts from middle-educated families to opt for postsecondary education, especially in contexts where tertiary education is more affordable. These findings suggest that recessions could increase social mobility in the middle strata of society if educational policies support affordability of tertiary education.

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Conflict of interest statement. None declared.

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Supplementary data

Supplementary data is available at *Socio-Economic Review* Journal online.

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